A PROJECT REPORT-
SAP INTEGRATION SUITE
END-TO-END PROCEDURE-TO-PAY
INTEGRATION DESIGN AND BLUEPRINT

Submitted to SAP Integration (IE), Industry 4.0

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Field	***Details***
Project Title	SAP Integration Suite: End-to-EndP2P Integration Blueprint
Course	SAP Integration Developer
Company (Fictitious)	TechMart Pvt. Ltd.
SAP Components	SAP BTP | SAP CPI | SAP MM | SAP FI
Integration Style	REST API s, i Flows, SFTP, Message Mapping

Title

SAP Integration Suite: End-to-End Procure-to-Pay (P2P) Design and Blueprint:

Integration

This project focuses on creating a system where SAP automatically sends purchase orders to vendors and gets invoices back. This will happen without any manual work and will use SAP Integration Suite on SAP BTP.

Problem Statement

At TechMart Pvt. Ltd., the procurement and finance departments operated in silos. Purchase Orders raised in SAP MM were manually emailed to vendors. Vendors replied with invoices via email or WhatsApp, which were then manually entered into SAP FI by the finance team. This manual process caused:

- Delays of 5–7 days per PO-to-payment cycle due to manual handoffs
- Data entry errors causing invoice mismatches and payment holds
- No real-time visibility into PO status or vendor delivery confirmations
- Duplicate invoice payments due to lack of automated three-way matching
- Compliance risks from missing audit trails across procurement and finance
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So the goal of this project is to design a efficient solution using SAP integration Suite that would automatically send purchase orders to the vendor and the invoices will come into SAP by itself without any human engagement. This would save time and and reduce human errors

Solution / Features

The proposed solution uses four i Flows deployed on SAP Integration Suite (SAP CPI) acting as a central middleware hub between SAP ECC (MM + FI modules) and the external Vendor Portal.

Core iFlows (Integration Flows)

iFlow	*Direction*	*Protocol*	*Key Feature*
iFlow 1: PO_Send_to_Vendor	SAP MM Vendor Portal	REST POST	IDOC JSON mapping, retry 3×
iFlow 2: Vendor_ACK_to_SAP	Vendor Portal SAP MM	REST GET	Poll every 5 min, update PO status
iFlow 3: Invoice_Receive	Vendor Portal SAP FI	REST POST	JSON XML, BAPI post, 3-way match
iFlow 4: Payment_Confirm	SAP FI Vendor Portal	REST PATCH	Payment cleared notification

Key Features of the Solution

- ***Automated PO Delivery:*** POs are sent to the vendor portal within seconds of creation in SAP MM, with no manual email needed.
- ***Real-Time Vendor Acknowledgement:*** iFlow 2 polls the vendor portal every 5 minutes and updates PO confirmation status in SAP automatically.
- ***Automated Three-Way Match:*** When the vendor submits an invoice, iFlow 3 triggers SAP FI's automatic three-way match (PO + GR + Invoice), reducing manual verification by ~80%.
- ***Zero Data Loss:*** Failed messages are queued and retried automatically — no integration failure causes lost data.
- ***Full Audit Trail:*** Every message exchange is logged in SAP CPI Message Monitor with 90-day retention for compliance.
- ***Secure by Design:*** OAuth 2.0 for vendor portal authentication, mTLS for SAP ECC connections, PGP encryption for financial payloads.

Screenshots

The following screenshots illustrate the SAP Integration Suite environment. Screenshots are captured from the SAP BTP Free Trial account at account.hanatrial.ondemand.com.

Screenshot 1 — SAP BTP Cockpit (Trial Account)

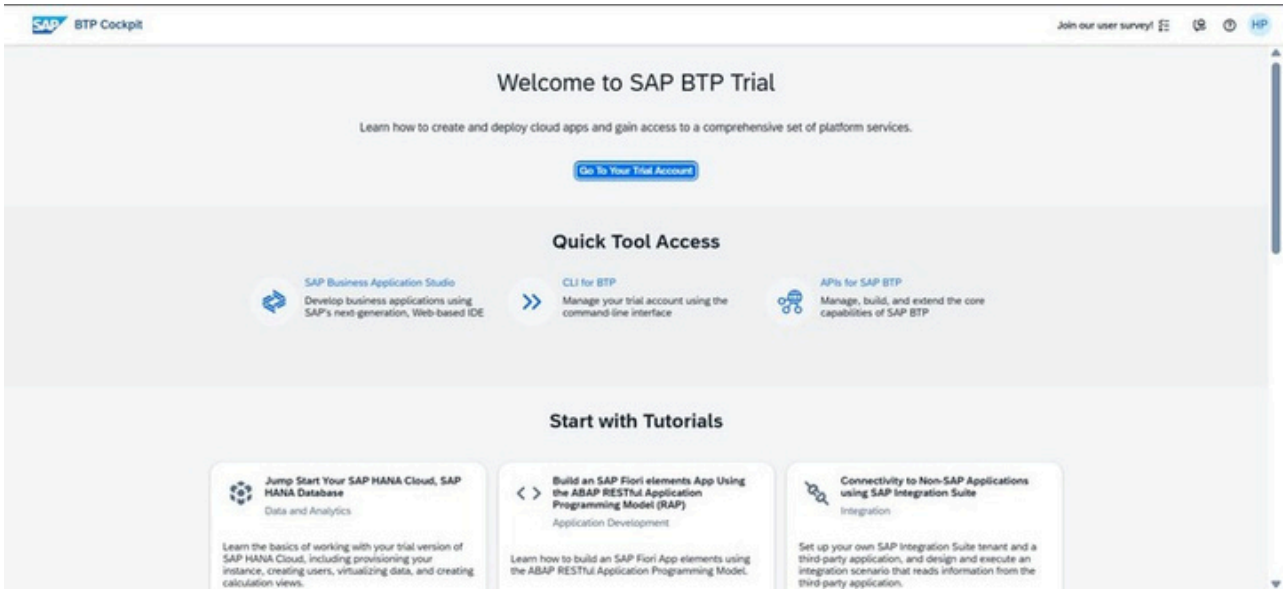

Figure 1: SAP BTP Trial Cockpit – Subaccount Overview

Screenshot 2 — SAP Integration Suite Landing Page

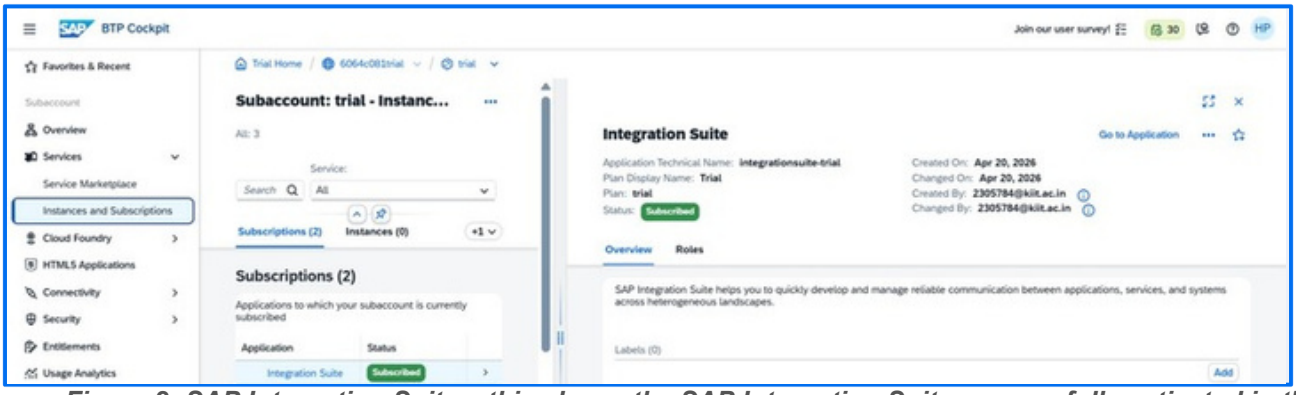

Figure 2: SAP Integration Suite – this shows the SAP Integration Suite successfully activated in the In the SAP BTP subaccount, which is acting as the middleware for integration

Screenshot 3 — iFlow Canvas

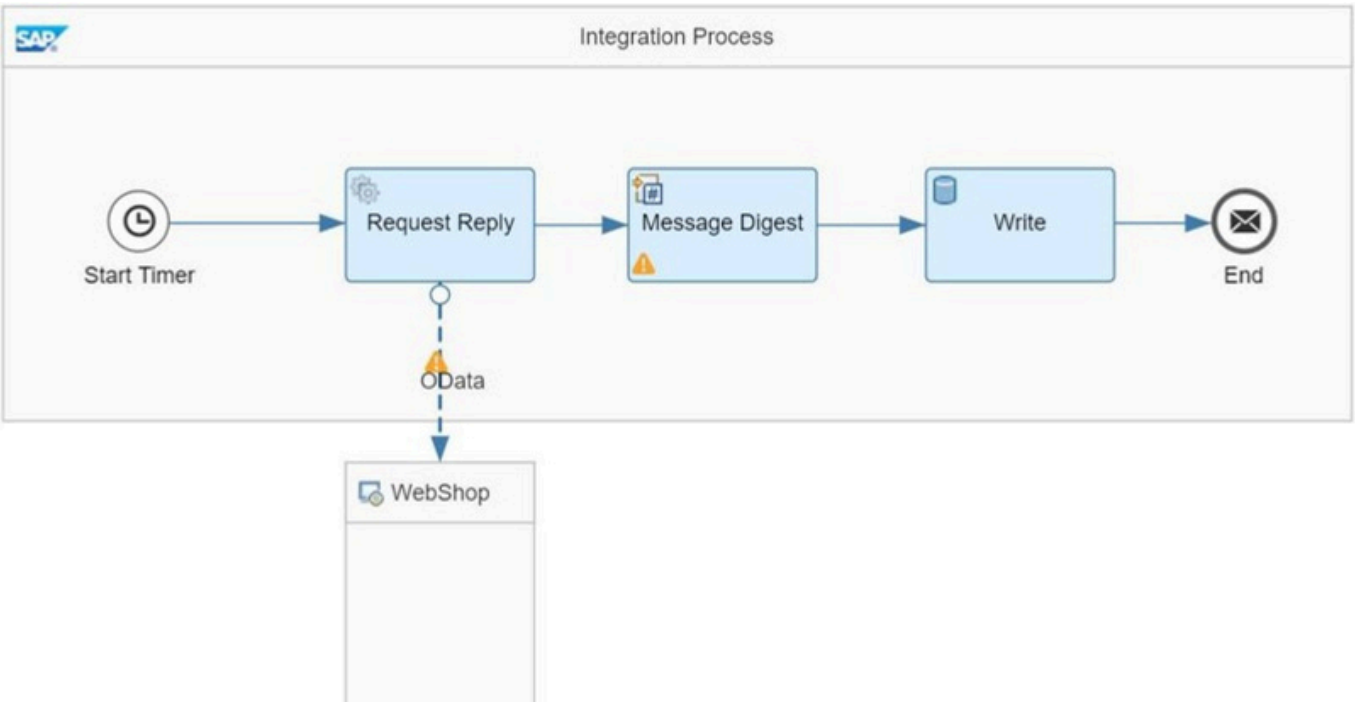

Figure 3: iFlow Canvas – PO_Send_to_Vendor Integration Flow

Screenshot 4 — Message Monitoring Dashboard

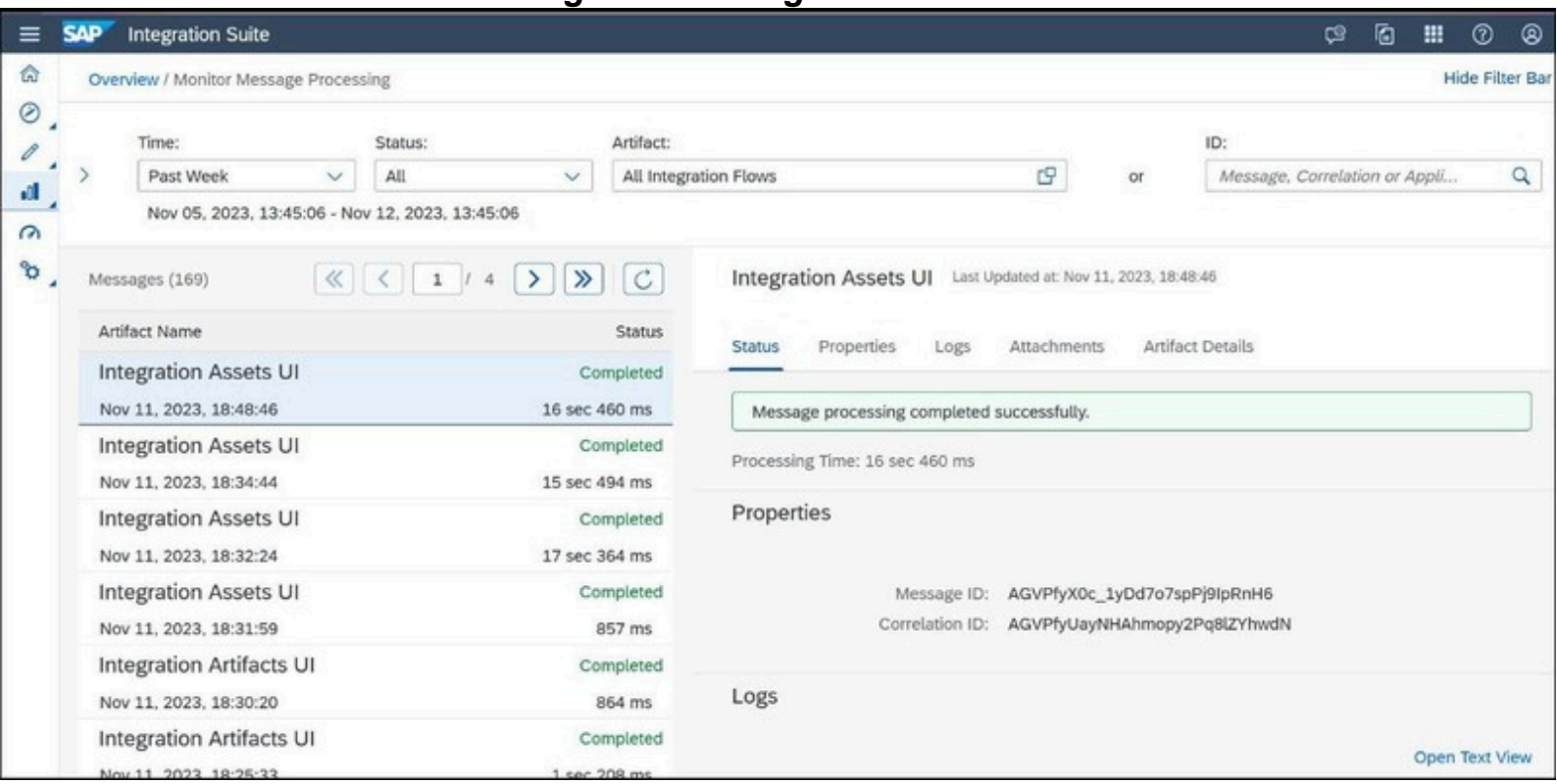

Figure 4: SAP CPI Message Monitoring – Message Processing Log
(taken from google as a example)

Tech Stack

Field	Details
SAP BTP	BusinessTechnology Platform – Cloud Foundry runtime environment
SAP Integration Suite	SAP CPI– iFlows, Message Mapping, Adapters, API Management
SAP ECC 6.0	Backend ERP – SAP MM (Materials Mgmt) + SAP FI (Finance)
Vendor Portal	External SaaS system – REST API, JSON format, HTTPS
Protocols	REST (HTTP/HTTPS), RFC/BAPI, SFTP, SOAP
Message Formats	XML (SAP IDOC ORDERS05), JSON, CSV (batch invoices)
Security	OAuth 2.0 Client Credentials, mTLS, PGP Encryption, X.509
Monitoring	SAP CPI Message Monitor, SAP Alert Management, Email Alerts
Design Tools	SAP BTP Cockpit, SAP Integration Suite Web UI, draw.io

Unique Points

- ***Idempotency Design:*** Every iFlow checks PO number / invoice number before processing to prevent duplicate entries — a critical requirement in financial integrations that mostbeginnerdesigns miss.
- ***Asynchronous Error Recovery:*** Failed messages are not discarded. They are queued andautomaticallyretried,ensuring zero data loss even during vendor portal outages.
- ***Three-Way Match Automation:*** The integration design triggers SAP FI's built-inthree-way matching (PO + GR+ Invoice) automatically upon invoice receipt, removingthe needfor manual finance verification.
- ***Centralized Credential Management:*** All API keys and passwords are stored inSAPCPISecureParameter Store— never hardcoded in any iFlow configuration.
- ***Scalable Architecture:*** TheiFlow design supports parallel processing for bulk invoice batches,allowingthe systemto handle 5× normal volume during month-end closing.

Future Improvements

Enhancement	***Description***	***Priority***
SAP Ariba Integration	Connect Ariba Procurement with SAP MM via pre-built CPI packages	High
Event-Driven via Event Mesh	Replace REST polling with SAP Event Mesh for real-time vendor ACK	High
AI Invoice Capture	Use SAP Document Information Extraction (DOX) for invoice OCR	Medium
Vendor Self-Service API	Expose PO status via SAP API Management for vendor self-service	Medium
SAP Analytics Cloud	P2P KPI dashboard: PO cycle time, GR accuracy, payment aging	Low

Conclusion

Before taking this course, I didn't really understand how different SAP systems communicated with each other. Through this project, I gained a clear understanding of how SAP Integration Suite works as a central middleware that connects SAP MM, SAP FI, and external systems smoothly. While designing the Procure-to-Pay (P2P) integration scenario for TechMart Pvt. Ltd., I learned how iFlows are structured and how they automate communication between systems.

One of the most valuable lessons for me was understanding data transformation. In real business situations, SAP often sends data in IDOC format, while external vendor systems need JSON. This project helped me see how message mapping within iFlows converts data between different formats, ensuring smooth communication. I also recognized the importance of error handling and security, which are vital for making integrations reliable.

More importantly, this project changed my perspective on integration. It is not just a technical task of connecting systems; it is a way to make business processes faster, more accurate, and easier to track. For example, by automating the P2P cycle, TechMart can cut its purchase-to-payment time from several days to less than 24 hours, while also reducing manual errors and improving visibility across departments.

Overall, this project gave me both clear concepts and practical insights into real-world SAP integration situations. It has strengthened my interest in becoming an SAP Integration Developer, and I look forward to using these skills in real industry projects.